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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction

Keeping organised and managing duties are essential for both individuals and teams in the fast-paced world of today. People continuously manage to juggle a multitude of duties, deadlines, and initiatives, whether in a personal, professional, or educational context. Good task management solutions are necessary to increase output, keep focus, and guarantee that assignments are completed on time. A well-thought-out Task Management Application facilitates priority setting, task organisation, teamwork, and progress monitoring.

In order to help users optimise their productivity, contemporary task management apps incorporate functions like project monitoring, notifications, time management, and collaboration capabilities in addition to offering a to-do list. The need for digital tools to organise and monitor projects has increased dramatically as remote work and dispersed teams have become more common.

1.1. Problem Scenario

The issue of effectively managing several projects, deadlines, and priorities faces many people as well as organisations. Typical problems include the following:

- Lack of organisation: It's challenging to remain on top of things because tasks are dispersed throughout several tools, notes, and platforms.
- Difficulty in tracking progress: Users may encounter difficulties determining the amount of work completed or the tasks that remain unfinished, which can result in missed deadlines or redundant efforts.
- No central hub for collaboration: Without a centralised platform for task assignment, tracking, and communication, it might be difficult for teams to work together efficiently.

- Limited visibility: Planning and resource allocation may be challenging in larger teams due to managers' and team leads' lack of awareness of the general status of tasks or projects.

These problems result in inefficiencies, missed deadlines, decreased productivity, and miscommunication within the team when a task management system isn't in place.

1.2. Project as Solution

The purpose of the Task Management Application is to address the aforementioned issues by offering a centralised platform that facilitates task management, prioritisation, and efficient teamwork. The app makes it easy for people to stay organised, monitor their progress, and fulfil deadlines. The following are salient characteristics that tackle the issues noted:

1. Centralised Task Management: – Users can establish, arrange, and rank tasks centrally. This eliminates the need for manual tracking or various tools.

2. Work Assignment & Teamwork: - In collaborative settings, responsibilities can be delegated to particular team members. Task descriptions, due dates, and priority rankings can be added to each one. Through comments or discussion threads inside each task, team members can work together to improve communication and decrease misunderstandings.

3. Task Prioritisation and Categorisation: To assist users in concentrating on the most crucial tasks, tasks can be prioritised (low, medium, high) and categorised using labels or tags (e.g., work, personal, urgent).

4. Progress Tracking: Using visual cues like progress bars or checklists, users can quickly evaluate task progress and track task status (e.g., pending, in-

process, completed). Users can remain cognisant of their progress and the tasks that remain unfinished thanks to this feature.

5. Notifications and Reminders: - When assignments are updated, have a deadline approaching, or are past due, users get automated reminders and notifications. By doing this, users can stay on course and lower their chance of missing deadlines.

6. Calendar Integration: - Users may view all tasks based on deadlines with the application's calendar view, which facilitates better schedule planning.

7. Project Management: - To help teams stay on task, the program groups tasks under specific projects and provides a dashboard that shows the project's overall status.

8. User Roles and Permissions: - Managers have further control over task assignments and tracking by configuring the app with various user roles (e.g., admin, manager, team member).

9. Reporting and Analytics: - To improve decision-making and performance analysis, users and team leads can create reports on task completion, time spent on tasks, and team productivity.

10. Mobile Accessibility: - Thanks to its responsive design, users can handle chores on the move by accessing the program from any device, be it a desktop, tablet, or mobile phone.

2. Aims and Objectives

2.1. Aims

The Task Management tool project aims to create an effective, user-friendly web tool that helps teams and individuals efficiently prioritise, track, and organise activities. With features including task assignment, real-time notifications, progress tracking, and task categorisation, the program will offer a centralised platform for task management, foster collaboration, and increase productivity.

2.2. Objectives

Objectives of the Project

1. To implement user authentication and authorization:

- Develop a secure login and registration system to allow users to create accounts and sign in.
- Implement role-based access control (e.g., Admin, User) to ensure that users have appropriate permissions to perform actions like creating, assigning, or deleting tasks.

2. To enable task creation, organization, and prioritization:

- Provide functionality to create, update, delete, and prioritize tasks based on importance (e.g., low, medium, high priority).
- Allow users to organize tasks using categories or tags (e.g., personal, work, urgent) for easy filtering and management.

3. To facilitate task assignment and collaboration:

- Implement features to assign tasks to specific users or teams, making collaboration easy in both personal and team environments.
- Enable task discussions and comments within tasks, allowing team members to collaborate, ask questions, and track task-related conversations in real-time.

4. To provide task progress tracking and visualization:

- Implement task status options (e.g., pending, in-progress, completed) to track the progress of each task.
- Add visual indicators such as progress bars, checklists, and Kanban boards to help users see how much work has been completed.

5. To integrate a notification and reminder system:

- Develop a real-time notification system to inform users of important task updates, including task assignments, changes in task status, and approaching deadlines.
- Implement email or push notifications for reminders on due dates or overdue tasks.

3. Expected Outcomes and Deliverables

3.1. Outcomes:

1. Enhanced Prioritisation and Task Organisation:
- By classifying tasks into distinct groups (work, personal, etc.), allocating priority levels, and establishing deadlines, users will be able to organise tasks effectively. Better task management and more productivity will result from this.
2. Improved Team Cooperation: The application will facilitate smooth teamwork by assigning tasks, providing real-time notifications, and facilitating task-specific discussions. Through a centralised platform, team members can communicate and monitor task progress, which will cut down on misunderstandings and redundant work.
3. Successful Tracking of Tasks and Progress:
- Visual tools like checklists, Kanban boards, and progress bars will allow users and managers to track the status of tasks and projects in real time. As a result, the workflow will be more transparent, making it possible for team members to spot bottlenecks and raise job completion rates.
4. Enhanced Productivity with Notifications and Reminders: - By including notifications and reminders for task deadlines and updates, users will be kept abreast of their obligations, resulting in a decrease in missed deadlines and an increase in task completion rates.
5. Recurring Task Management and Task Dependency:
The ability to establish task dependencies and generate repeating tasks will be helpful to users as it will guarantee that repetitive jobs are automated and that tasks are executed in the proper order.

6. Data-Driven Analytics-Based Decision Making:

- Users and managers will be able to evaluate task completion rates, time spent on tasks, and overall productivity through reports and analytics produced by the program. To improve processes, well-informed decisions can be made using this data.

7. Cross-Device Accessibility: - Users will be able to manage their tasks while on the go thanks to the mobile-responsive design, which will also guarantee that tasks can be accessed and modified from any location and on any device.

3.2. Deliverables:

1. Online-Based Task Management Application:

- A completely functional online application with all essential features (task creation, task assignment, task progress monitoring, etc.) that was constructed with React for the frontend and .NET for the backend.

2. Role-Based Access Control and User Authentication:

- An installed secure user login system that separates users into three roles (Admin, Manager, and Team Member) so that users may only access features and tasks that correspond to their rights.

3. Features for Task Management and Collaboration:

- Task creation, editing, deletion, and assignment functionality; includes teamwork features like file attachments and task-specific comments.

4. Progress Tracking and Visualization: - A dashboard that enables users to efficiently manage tasks and projects by providing tools for visualising work progress through progress bars, Kanban boards, and other visual components.

5. Real-Time Notifications and Reminder System:

- A fully integrated notification system that sends alerts for important task updates, deadlines, and assignment changes, including email and/or push notifications.

6. Subtasks and Task Dependencies:

- Implementation of subtasks under larger tasks and task dependencies to ensure tasks are completed in the correct sequence.

7. Calendar View for Task Scheduling:

- A calendar feature where users can see upcoming tasks and deadlines, allowing them to schedule their time more effectively.

8. Task Performance Reports and Analytics:

- A reporting feature that generates analytics on task performance, completion rates, time spent on tasks, and other key productivity metrics.

9. Mobile-Friendly Design That Responds:

- A responsive web interface that guarantees the program runs without hiccups on laptops, tablets, and mobile phones, offering a seamless experience on all platforms.

10. Recordkeeping:

- Extensive documentation, comprising:
 - User manual: End-user instructions for system operation.
 - Technical documentation: Information on the database design, codebase, and system architecture.
 - API documentation (should any built APIs or external integrations be made).

11. Reports on Testing and Debugging:

- Results from user acceptability testing, functional testing, and unit testing to make that the program is error-free and performs to performance requirements.

12. Final Presentation and Demonstration:

- A live demonstration of the program together with a final presentation outlining its main features, the issue it resolves, and the advantages it offers users.

4. Methodology:

4.1. Agile Methodology:

The development of the Task Management Application will follow a structured methodology that encompasses several phases, ensuring systematic progress from initial planning to final deployment. The chosen methodology is Agile, which allows for flexibility, iterative development, and continuous feedback.

1. Requirement Analysis

- Objective: Gather and analyze user requirements to understand the needs and expectations for the application.
- Activities:
 - Conduct interviews and surveys with potential users to identify their pain points and desired features.
 - Create use cases and user stories to define how users will interact with the application.

2. Planning

- Objective: Define the project scope, timeline, and resources needed for development.
- Activities:
 - Establish project milestones and deliverables.
 - Create a project schedule using tools like Gantt charts or Agile boards.
 - Assemble the development team and assign roles and responsibilities.

3. Design

- Objective: Develop the architectural and user interface design of the application.
- Activities:

- Design the application architecture, including the backend (using .NET) and frontend (using React).
- Create wireframes and mockups for the user interface, focusing on user experience (UX) and accessibility.
- Define the database schema for storing tasks, user information, and related data.

4. Development

- Objective: Build the application iteratively, implementing features in small, manageable increments.
- Activities:
 - Set up the development environment and version control (e.g., Git).
 - Implement core features according to user stories, such as task creation, assignment, and progress tracking.
 - Integrate additional features incrementally, ensuring each feature is tested before moving on to the next.
 - Conduct regular code reviews to maintain code quality and consistency.

5. Testing

- Objective: Ensure the application functions as intended and is free of bugs.
- Activities:
 - Perform unit testing for individual components and features.
 - Conduct integration testing to verify that different parts of the application work together correctly.
 - Execute user acceptance testing (UAT) with a group of end-users to gather feedback and identify any issues.
 - Document test cases and results for reference.

6. Deployment

- Objective: Launch the application to a production environment for end-users.
- Activities:
 - Prepare the server environment for hosting the application (e.g., cloud hosting, web server setup).

- Deploy the application and perform final checks to ensure everything is functioning correctly.

- Provide user training and documentation to facilitate adoption.

7. Maintenance and Support

- Objective: Provide ongoing support and updates to ensure the application remains functional and relevant.

- Activities:

- Monitor the application for bugs, performance issues, and user feedback.

- Implement regular updates to fix bugs, improve features, and add new functionalities based on user input.

- Offer support channels for users to report issues or seek help.

8. Evaluation and Feedback

- Objective: Assess the project outcomes against the initial objectives and gather user feedback for future improvements.

- Activities:

- Conduct post-launch surveys or interviews with users to evaluate their experience.

- Analyze usage data to identify areas for enhancement or additional features.

- Document lessons learned and recommendations for future projects.

5. Software and Hardware Requirements

5.1. Software Requirements

1. Backend Development:

- Framework: .NET (preferably .NET Core or ASP.NET)

- Database Management System: SQL Server or MySQL

- Development Tools: Visual Studio or Visual Studio Code

2. Frontend Development:

- Framework: React.js
- Package Manager: npm (Node Package Manager)
- UI Library (optional): Material-UI, Ant Design, or Bootstrap for styling

3. Version Control:

- Tool: Git (with platforms like GitHub or GitLab for repository hosting)

4. Testing Tools:

- API Testing: Postman for testing backend APIs

5. Project Management and Collaboration:

- Trello for task management

6. Design Tools:

- UI/UX Design: Figma for creating wireframes and mockups

5.2. Hardware Requirements

1. Development Machine:

- Processor: Intel i5 or equivalent
- RAM: Minimum 8 GB (16 GB recommended for better performance)
- Storage: SSD with at least 256 GB of free space

2. Testing Devices:

- Devices for Testing: Various devices (smartphones, tablets, desktops) with different operating systems and browsers to ensure cross-device compatibility and responsiveness.

6. Conclusion:

The Task Management Application offers a feature-rich, user-friendly platform for task management and tracking, which effectively addresses the major issues of work disorganisation, inefficiency, and lack of collaboration. With real-time collaboration, notifications, and progress tracking, the software guarantees that deadlines are fulfilled, boosts productivity, and assists individuals and groups in staying organised. The increasing demand for efficient task management in social, academic, and professional contexts is met by this approach.